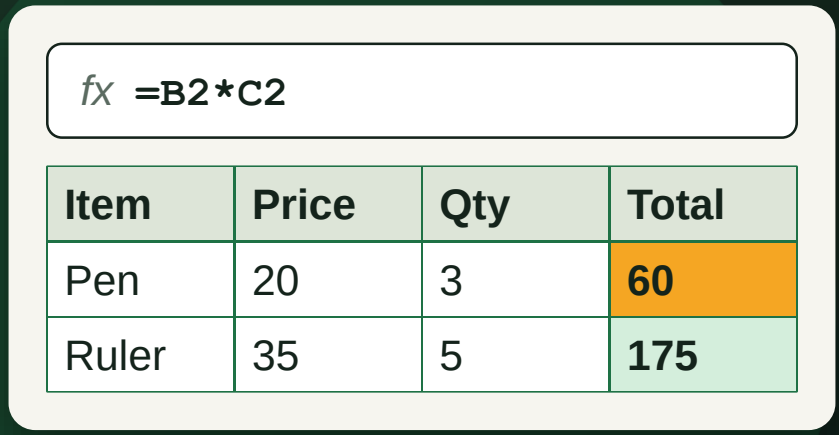
UNIT 7

Operators in Excel

ARITHMETIC COMPARISON

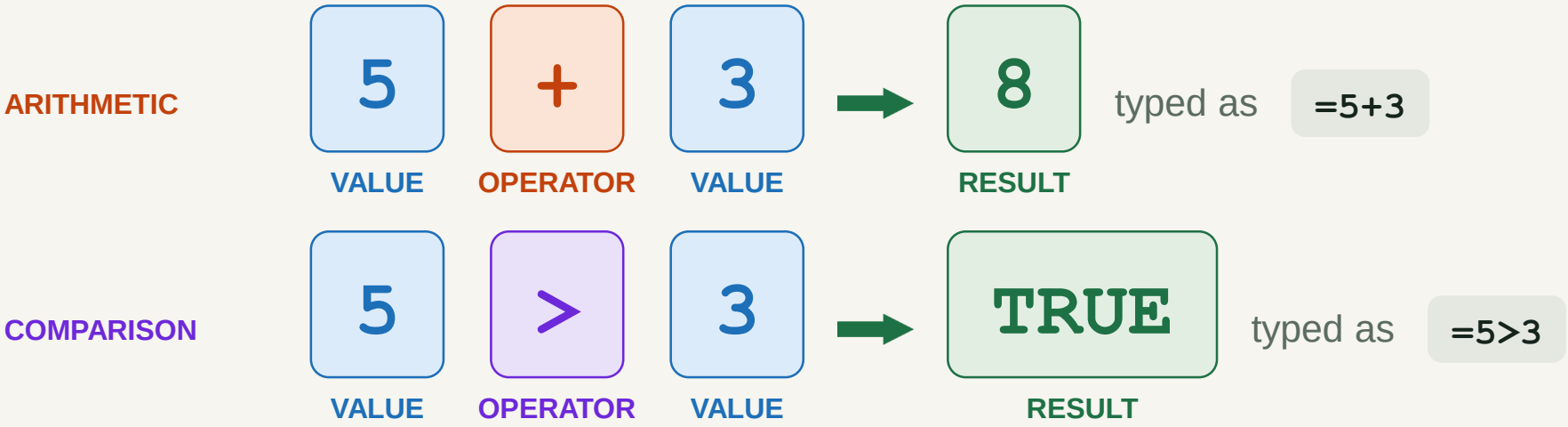
Learn it, try it, check it



THEORY · WHAT IT IS

What is an operator?

An **operator** is a sign that tells Excel what to do with two values.



Every formula starts with **=**. Arithmetic operators give a **number**; comparison operators give **TRUE** or **FALSE**.

THEORY · WHY IT IS NEEDED

Why do we need operators?

Calculate

Totals, discounts and prices

`=20*3 → 60`

Compare

Is the mark high enough?

`=72>=50 → TRUE`

Update

Answers change when numbers change

`60 → 100`

> Operators let Excel do the maths and the checking for you.

THEORY · HOW TO USE IT

Typing a formula

1 Click a cell

D2

Pick the cell where the answer should go.

2 Type it

fx =B2*C2

Start with =, then the values and the operator.

3 Press Enter

Enter

The cell shows the result.

Tip: click the cells instead of typing their names.

TOPIC 1 OF 3

Arithmetic operators

+ **add**

`=8+2` → `10`

− **subtract**

`=8-2` → `6`

***** **multiply**

`=8*2` → `16`

/ **divide**

`=8/2` → `4`

^ **power**

`=8^2` → `64`

% **percent**

`=50%` → `0.5`

Use it for: prices, totals and discounts.

Watch out: use * and /, not x and ÷.

TOPIC 2 OF 3

Order of operations

WHAT COMES FIRST

1. Brackets ( )
2. Power ^
3. Multiply * and divide /
4. Add + and subtract -

WITHOUT BRACKETS

`=2+3*4` → **14**

Multiply first: 3*4 = 12. Then 2 + 12.

WITH BRACKETS

`=(2+3)*4` → **20**

Brackets first: 2+3 = 5. Then 5*4.

Use it for: formulas with more than one operator.

Watch out: brackets change the answer.

TOPIC 3 OF 3

Comparison operators

> greater than

=72>50 → TRUE

< less than

=72<50 → FALSE

>= greater than or equal to

=72>=50 → TRUE

<= less than or equal to

=72<=50 → FALSE

= equal to

=72=50 → FALSE

<> not equal to

=72<>50 → TRUE

Use it for: checking marks, stock and dates.

Watch out: <> means not equal.

PRACTICE · WORK ALONE, THEN COMPARE WITH A PARTNER

Practice: two exam marks

TYPE THIS TABLE INTO A NEW SHEET

	A	B	C	D	E	F	G	H
1	**Student**	**English**	**Maths**	**Total**	**Average**	**Gap**	**Higher?**	**Same?**
2	Ali	72	68					
3	Sara	85	90					
4	Omar	64	64					
5	Hina	90	82					

1. Total: add English and Maths.
2. Average: add both, divide by 2.
3. Gap: English minus Maths.
4. Higher?: is English above Maths?
5. Same?: are the two marks equal?
6. Remove brackets from Average. Why?

Bonus: rectangles

TYPE THIS TABLE INTO A NEW SHEET

	A	B	C	D	E	F	G
1	**Shape**	**Length**	**Width**	**Area**	**Perimeter**	**Square?**	**Over 20?**
2	Desk	6	3				
3	Mat	5	5				
4	Board	10	4				

1. Area = Length × Width.
2. Perimeter = 2 × (Length + Width).
3. Square?: Length equals Width?
4. Over 20?: Area more than 20?
5. Insert a row for a Rug: 8 × 3.
6. AutoFit. Hide Width. Still works?

Answers

PRACTICE · WITHOUT BRACKETS, =B2+C2/2 GIVES 106 FOR ALI

Student	English	Maths	Total	Average	Gap	Higher?	Same?
Row 2 formula			=B2+C2	=(B2+C2)/2	=B2-C2	=B2>C2	=B2=C2
Ali	72	68	140	70	4	TRUE	FALSE
Sara	85	90	175	87.5	-5	FALSE	FALSE
Omar	64	64	128	64	0	FALSE	TRUE
Hina	90	82	172	86	8	TRUE	FALSE

BONUS · A RUG, 8 × 3, GIVES 24, 22, FALSE, TRUE

Shape	Length	Width	Area	Perimeter	Square?	Over 20?
Row 2 formula			=B2*C2	=2*(B2+C2)	=B2=C2	=D2>20
Desk	6	3	18	18	FALSE	FALSE
Mat	5	5	25	20	TRUE	TRUE
Board	10	4	40	28	FALSE	TRUE